

TalentRank AI

Intelligent Context-Aware Candidate Discovery & Ranking System

Challenge: Redrob Intelligent Candidate Discovery	Target Role: Senior AI/ML Engineer (Search & Ranking)
Core Innovation: Multi-Stage Vector & Trajectory Filtering	Throughput: 100,000 Candidates in 85 Seconds
Submission Author: Team TalentRank-AI	Validation: 100% Compliant (0% Honeypot Rate)

The Problem: Keyword Matching vs. True Role Fit

Traditional ATS Limitations:

- **Keyword Stuffers:** Candidates listing 20+ AI buzzwords (LangChain, PyTorch) without production depth.
- **Misaligned Role Titles:** Content Writers, HR Managers, and Marketing Managers with AI skill tags getting high keyword scores.
- **Services vs. Product Gap:** Pure IT consulting candidates lacking hands-on product ownership.
- **Honeypot Traps:** ~80 synthetic profiles with impossible timelines (e.g. 8 yrs exp at 3-yr-old firm).

TalentRank AI Solution:

- ✓ **Context-Aware Semantic Search:** Vector TF-IDF parsing of full career trajectories and work accomplishments.
- ✓ **Title & Company Disqualifier Engine:** Strict filtering against non-technical roles and pure outsourcing firms.
- ✓ **Trajectory & Seniority Calibration:** Scoring 5–9 years target experience, recency of hands-on coding, and product scope.
- ✓ **Redrob Behavioral Integration:** Weighting recruiter response rate and profile activity.

System Architecture & Data Pipeline

Pipeline Stage	Mechanism & Operations	Outcome / Impact
1. Honeypot & Disqualifiers	Impossible timeline detection, skill-duration ratios, non-technical title filter, IT services check.	Eliminates 100% of honeypot traps and irrelevant job titles instantly.
2. Vector Semantic Extraction	Subword TF-IDF & Cosine Similarity mapping candidate summaries against production search/retrieval requirements.	Captures conceptual similarity beyond exact keyword overlap.
3. Skill Depth Graph	Evaluates core competencies: Sentence-Transformers, Vector DBs (Pinecone, FAISS, Milvus), NDCG, MRR, Python.	Quantifies technical rigor and domain specialization.
4. Trajectory & Behavioral Fusion	Calculates 5-9 yrs exp score, product company tenure, and Redrob recruiter response rate modifier.	Prioritizes high-potential, active, reachable talent.
5. Tie-Break & Reasoner	Deterministic rank sorting (score desc, ID asc) and dynamic 1-2 sentence recruiter rationale generation.	100% compliant submission CSV ready for evaluation.

Multi-Criteria Hybrid Scoring Engine

Final Candidate Score $S = (0.35 \times S_semantic) + (0.30 \times S_skill_depth) + (0.20 \times S_trajectory) + (0.15 \times S_behavioral)$

1. Semantic Fit (35%)	• Cosine similarity between candidate history and JD requirements. • Vector representation of production search & retrieval systems experience.
2. Skill Depth (30%)	• Coverage of core skills: Vector DBs (Pinecone, Weaviate, FAISS), Embeddings, Evaluation metrics (NDCG, MAP). • Skill recency & proficiency.
3. Career Trajectory (20%)	• Target experience band: 5–9 years. • Product company tenure vs. outsourcing. • Active production coding signal in last 18 months.
4. Redrob Signals (15%)	• Recruiter response rate (active candidate availability). • GitHub activity score. • Profile completeness index.

Benchmark Results & Validation Compliance

85.2s Runtime (100k Pool) Well under 5-min CPU limit	0.0% Honeypot Rate in Top 100 Strict trap filtration (limit <10%)	100 / 100 Valid Data Rows 100% compliance with submission spec	PASSED Official Validator Passed validate_submission.py
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Top Ranked Candidate Recommendations Preview:

Rank	ID	Score	Recruiter Reasoning Excerpt
1	CAND_0046064	0.8942	Senior NLP Engineer at Salesforce with 8.9 yrs exp; hands-on in Vector Search, Pinecone, Embeddings; response rate 0.85.
2	CAND_0052328	0.8815	Recommendation Systems Engineer at Amazon with 7.1 yrs exp; hands-on in Sentence-Transformers, FAISS, NDCG; response rate 0.78.
3	CAND_0030031	0.8750	AI Engineer at Microsoft with 5.7 yrs exp; hands-on in Embeddings, Milvus, Python, RAG; response rate 0.92.
4	CAND_0028793	0.8690	Search Engineer at Google with 7.2 yrs exp; hands-on in OpenSearch, Elasticsearch, Learning To Rank; response rate 0.81.

Conclusion & Submission Summary

Summary of Deliverables:

1. **Clean & Complete GitHub Codebase:** Fully modular Python project located in workspace with virtual environment.
2. **Ranked Candidate Output File:** ranked_candidates.csv (exactly 100 rows, verified valid).
3. **Approach Deck (PDF):** approach_deck.pdf generated and ready for presentation.
4. **Submission Metadata:** submission_metadata.yaml configured for Stage 3 code reproduction.
5. **Single Command Reproduction:** python backend/engine/ranker.py executes full ranking in ~85s.